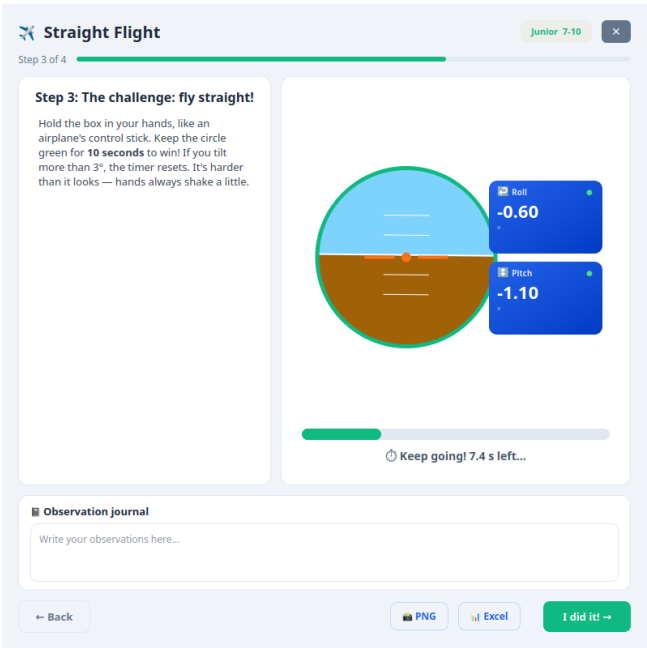


Flying Straight

The box becomes a spirit level and an artificial horizon — and ten seconds of stillness turn out to be surprisingly hard

Level: Junior — ages 7–10 (Romanian school years II–IV) • **Time in class:** 15 min, or one 45-minute lesson • **Sensors:** MPU-6500 (acceleration)



The activity screen. The values in the picture are simulated; your own will differ.

Before you start

Who it is for	Junior — ages 7–10 (Romanian school years II–IV). The numbers are small and signed — -7° , $+2^\circ$ — but a child does not need to read them in order to play: the circle on screen turns green or it does not. The numbers come into the discussion from year III, and their sign can be handled simply, as “one way” and “the other way”.
Assumed knowledge	Nothing compulsory. Comparing two small numbers helps, but the challenge can be done watching only the colour of the circle. Angles as a syllabus topic are not required.
Time in class	Short: 15 min — the wizard’s four steps. Full: one 45-minute lesson, with the tour of surfaces in the room and the discussion about pilots’ instruments.
Teacher preparation	5 min — find one surface you know to be level and one you know is not; and try the challenge yourself once, so that you know how hard it is (it is harder than it looks).
Sensor settling time	None. The accelerometer responds instantly. The box must, however, rest <i>on a surface</i> , not in a palm — no hand, however steady, holds three degrees.

Sensors and boxes

MPU-6500. **The box is mounted upside down and the firmware already corrects for it:** a box lying flat reports $AZ \approx +1$ g. A box reading -1 g flat is running old firmware and needs reflashing. One box is enough for the whole class with the screen projected; ideally, one per 4–6 children.

Three routes**Short**

15 min: the four steps — what roll and pitch are, the digital level, the ten-second challenge, and why pilots need instruments.

Full

The 45-minute lesson: the short route plus **the tour of surfaces** — the desk, the window sill, the floor, the top of a cupboard, a book placed underneath — and ranking them by how level they are.

Project

Extension: the challenge with eyes closed for whoever holds the box, with a classmate saying only “higher” or “lower” — which is exactly what an instrument does for a pilot who cannot see outside.

1. Where this fits in the curriculum

- **Mathematics and exploring the environment** (years I–II) — orientation in space and an object’s positions: tilted one way, tilted the other, level. **The words “horizontal” and “tilted” gain an instrument that checks them**, which is rare at this age.
- **Natural sciences** (years III–IV) — balance and the forces acting on a body; and, more simply, the idea that an instrument can know something the eye cannot: whether a table really is level.
- **Mathematics** (from year III) — comparing small numbers and the idea of a *tolerance*: not “zero” but “smaller than three”. This is a very mature idea, met here in a form a nine-year-old can use.
- **Physical education** — fine control of the hand and quiet breathing; ten seconds of stillness is a real test of motor control.
- **Cross-curricular** — why aircraft have instruments, and what happens when a pilot enters cloud. **This is the question that stays with children**, and the app’s last step asks it.

The core skill is **trusting an instrument when the senses mislead**. A child who holds the box “level” and sees 5° on screen has, for the first time, proof that their own judgement is not enough — and that is exactly a pilot’s situation in cloud.

2. Time plan

Route	Time	What happens
Short	15 min	Written prediction (2), what roll and pitch are (3), the digital level and two surfaces tested (4), the ten-second challenge (4), why pilots need instruments (2). These are exactly the wizard’s four steps.
Full	45 min	The short route unhurried (15), the tour of surfaces with five or six marked places (15), the challenge for every child in turn (10), the debrief (5).
Extension	+10 min	The “in cloud” challenge: whoever holds the box closes their eyes, and a classmate says only “higher” or “lower”.

Ten seconds are harder than they look — and that is the point

The challenge requires **both roll and pitch** to stay below **three degrees** for **ten seconds**, and the timer *resets* at the first departure. Try it yourself before the lesson: most adults need several attempts.

Three things make it possible, and they are worth saying to the children at the start: the box rests *on a surface*, not in a palm — a hand does not hold three degrees; the surface itself has to be level, or the challenge is impossible however well you hold it; and hands come off the table, because a table pressed at one corner tilts.

Failing the first attempts is the good part. A child who succeeds first time has learned nothing; one who succeeds on the fourth has discovered in the meantime what needed changing — and that is called fine adjustment.

3. Materials

Item	Per group	Notes
Magic Box with MPU-6500	1	One box and the projected screen works very well for a whole class; ideally, one per 4–6 children.
Laptop with the app	1	USB or WiFi. On a cable, the cable pulls the box — leave it slack or work over WiFi.
Surfaces to test	4–6	The desk, the sill, the floor, the top of a cupboard, a shelf. Marked with labels so that every group tests the same ones.
A book or two	1–2	Placed under one edge to make a deliberately tilted surface — the activity's control case.
A spirit level	1 (optional)	If the school has one, comparing with it is very convincing: two independent instruments saying the same thing.
The class clock	1	The app times the challenge itself, but a visible clock helps at the debrief.

Preparation. Find in advance **one surface that is level** and **one that is not** — window sills usually have a slight slope, put there deliberately so that water runs off, and that is an excellent discovery. Mark the places with labels. And try the challenge once, so that you know what to expect.

4. Scientific background for the teacher

How a box with no windows knows it is tilted

The accelerometer does not sense tilt. It senses **acceleration**, and at rest the only acceleration present is gravity: a vector pointing always downwards, of magnitude 1 g.

When the box is level, all of that vector falls on the vertical axis: $a_z = 1$ g, and a_x and a_y are zero. When the box tilts, the vector stays where it is — gravity does not tilt — but *the box's axes* rotate relative to it, so part of it appears on the horizontal axes. From the ratio of those components the angles are recovered:

$$\text{roll} = \arctan \frac{a_y}{\sqrt{a_x^2 + a_z^2}}, \quad \text{pitch} = \arctan \frac{-a_x}{\sqrt{a_y^2 + a_z^2}}.$$

This is exactly the idea of a spirit level, only with the bubble replaced by a measurement. And it is the idea of an aircraft's *artificial horizon* — the instrument that shows the pilot where the horizon is when they cannot see it.

Two notes for the teacher. First, the method works **only at rest**: if you move the box, the acceleration of the movement adds to gravity and the angles become false — which is why a real aircraft also uses gyroscopes. Second, in the formulas above a_z **enters squared**, so its sign does not matter; that is why this activity was unaffected by the change of mounting convention in the firmware, unlike others.

4.1 Why three degrees, and why ten seconds

Three degrees is a well-chosen tolerance: tight enough that it cannot be hit by accident, wide enough that an eight-year-old can reach it after a few attempts. It is worth showing how little it is: over the 50 cm length of a desk, three degrees means one end raised by about 2.6 cm — the thickness of a finger.

Ten seconds is far longer than it sounds. The human hand has a natural tremor, and breathing moves the whole body; that is why the box has to rest on a surface rather than in a palm. **The timer resetting at the first departure** is what turns the challenge from luck into a skill: it is not enough to hit the target, you have to *stay* there.

4.2 Why pilots become lost in cloud

The app's last step touches a real and serious subject. With no view outside, the human inner ear **cannot distinguish** a steady bank from level flight: after a few tens of seconds of a gentle turn, the sensation of banking fades, and the pilot “feels” level while actually turning.

The phenomenon is called spatial disorientation and it has caused well-known accidents. The instrument — the artificial horizon — is the only thing that contradicts the pilot, and instrument flight training is largely a matter of learning **to believe the instrument against your own sensation**.

For year IV, the right phrasing is simple and true: *sometimes your body lies to you, and the instrument does not*. The eyes-closed extension makes exactly that tangible, safely, in thirty seconds.

5. Lesson flow

The in-app wizard has **four steps**.

1. **The written prediction (2 min)**. Before anything: *do you think your desk is level? What about the window sill?* And then the good question: *how long do you think you can hold the box still — one second, ten, a minute?* Almost every child overestimates.
2. **What roll and pitch are (3 min)**. The app's first step. Show it with your own palm: tilting sideways is *roll*, tilting forward and back is *pitch*. The button is **Got it**.

Technical note — two names, two numbers and a circle that turns green

The sensor. MPU-6500. **The box is mounted upside down and the firmware already corrects it** — a flat box gives $AZ \approx +1$ g. **Roll and pitch do not depend on the sign of AZ** , because it enters the formulas squared.

How it starts — and a mismatch of names. The activity opens from the **Activities** page, from the **Flying Straight** card. **The window that opens, however, is titled *Straight Flight***. It is the same activity.

The two cards. Roll and Pitch, both in degrees, with a sign: negative one way, positive the other. Below them is the **artificial horizon**, with pitch reference marks every ten degrees.

The challenge, exactly. The circle turns green when **both** |roll| and |pitch| are below 3°. It has to be held green for **ten seconds**. **At the first departure from tolerance the timer resets to zero** — it does not resume where it left off. This is the rule that makes the challenge hard, and it is worth stating in advance so that failure does not look like a fault.

The box rests on a surface. A box held in a palm cannot stay under three degrees for ten seconds — hand tremor and breathing are larger than that. **This is not a weakness of the child**, and saying so beforehand saves a great deal of frustration.

The cable matters. A taut USB cable pulls the box and tilts it by a few degrees, steadily. Leave it slack on the table or work over WiFi — otherwise a perfectly level surface will look crooked.

The buttons. The green button changes its label at every step: **Got it, I tested the surfaces, I did it!** and, at the end, **Finish**.

Exporting the data. At the measurement steps, **PNG** and **Excel** appear. The acceleration data are in a separate file with the `_imu` suffix. **For this activity the export is rarely needed**: the result is the number of attempts, recorded on the sheet.

3. **The digital level — the tour of surfaces (4 min).** The box is placed in turn on the marked surfaces and roll and pitch are recorded. *Deliberately* put a book under one edge of one of them, so that there is a case you know to be crooked. The question: *which surface in this room is the most level?* The button is **I tested the surfaces**.
4. **The challenge (4 min).** Every child in turn tries to hold the circle green for ten seconds. **Count the attempts**; do not time who is fastest. The button is **I did it!**
5. **Why pilots need instruments (2 min).** The final step. Ask the question that ties it together: *if a pilot can see nothing through the window, how do they know they are flying level?* Close with **Finish**.

Debrief: the score is not the point

Do not ask “who did it first”. Ask “**on which attempt did you succeed, and what did you change between them?**”

That is the only question that turns the game into a lesson, and it has real answers that children find for themselves: *I put the box on the desk instead of holding it; I took my hands off the table; I moved the box towards the middle of the desk; I put a piece of paper under one corner; I kept still and breathed more slowly.* Every one of them is a **deliberate intervention followed by a check** — that is, scientific method, at eight.

Three debrief questions, in order:

- *What stopped you the first time?* — almost always the hand, the pressed table, or the cable.
- *What did you change, and did it work?* — this is where the intervention gets named.
- *If you tried on the window sill now, would you succeed?* — this tests whether they have grasped that the surface matters, not only the hand.

The score is not the objective. A child who succeeded on the sixth attempt and can say what they changed each time has learned more than one who hit it first time. Say so explicitly to the class *before* the challenge — otherwise the number of attempts becomes an embarrassment instead of a result.

The investigation, in one table

The question. Which surface in our room is the most level?

The hypothesis. Written on the sheet **before** any measuring: rank the four surfaces by how level you think they are. Not changed afterwards.

What I change. One thing only: the **surface** the box stands on.

What I measure. Roll and pitch, in degrees, with their signs.

What I keep the same. The same box, **the same orientation** — with the same edge towards me every time, because otherwise roll and pitch swap over — **the same place on the surface** (the middle, not the edge), and the cable left slack.

How many repeats. Two placings on each surface, lifting the box in between. If the two readings differ a lot, the box was not put in the same place.

What makes a reading valid — and what makes it an artefact. Valid is a reading taken with the box set down, hands off the table and the cable slack. An artefact is anything else: **a hand leaning on the table presses it out of level**, a taut cable pulls the box steadily to one side, and a box on the edge of a desk reads differently from the same desk at the middle. **If a surface gives the same angle however you rotate the box, then the surface really is tilted** — and that is the decisive test.

6. Analysis after the activity

6.1 What the data should look like

Situation	Typical value	What it means
A good desk, box in the middle	0–1°	Level. The challenge is possible here.
The same desk, box on the edge	1–3°	The top sags a little; choose the middle.
Window sill	1–5°	Often tilted on purpose , so that water runs off. A very good discovery.
Floor	0–2°	Usually the most level surface in the room.
With a book under one edge	5–15°	The control case: we know it is crooked.
Box held in a palm, held still	2–8°, trembling	Which is why the challenge needs a surface.
Box on a table with somebody leaning on it	shifts by 1–4°	The table tilts under the weight.
Taut USB cable	a steady offset of 1–3°	It pulls the box always the same way.

6.2 Questions to ask afterwards

1. You thought your desk was level. What did the instrument say? Who was right — your eyes or the box?
2. Why can you not hold the box in your hand for the challenge? What does your hand do even when you think it is still?
3. The window sill came out tilted. **Is that a builder's mistake?** What might a slight slope there be for?
4. When somebody leaned on the table the numbers changed. What does that tell us about how

“solid” a table is?

5. A pilot entering cloud can no longer see the horizon. Why can they not rely on what they feel, and what do they use instead?

6.3 Assessment ideas

- **Low stakes:** the child ranks the surfaces from most level to most tilted, using only the colour of the circle or the size of the numbers, without writing them down.
- **Standard:** the child records roll and pitch for each surface, compares them with the prediction, and says which surface surprised them.
- **Extension:** the child explains in writing why they only succeeded at the challenge after several attempts and what exactly they changed — that is, describes an intervention and its result.

Common pitfalls and misconceptions

The box held in a palm. Hand tremor is larger than three degrees. The challenge becomes impossible and the child thinks it is their fault. Say it first: **the box rests on a surface.**

Hands on the table. A table pressed at one corner tilts by a few degrees. For the challenge, everybody takes their hands off the table.

A taut cable. It pulls the box steadily and makes a level surface look crooked. The cable lies slack, or you work over WiFi.

Rotating the box between measurements. Turn it by 90° and roll and pitch swap over, ruining the comparison. **The same edge towards you** every time.

Measuring on the edge of the desk. Desk tops sag towards the edges. Measure at the middle, always in the same place.

Expecting zero to mean exactly zero. The last decimals always tremble. The three-degree tolerance exists precisely because perfect zero cannot be reached.

Turning the challenge into a speed contest. The number of attempts is a result, not an embarrassment, and whoever succeeds first time has learned nothing. See the debrief box.

Safety and classroom management

The box is not held over a hard floor unsupervised. It is small and slippery; for the challenge it is set on the table, not carried about the room.

Nobody climbs on chairs or desks to reach a high surface — the top of a cupboard is measured only if it can be reached standing on the floor, and otherwise the teacher measures it.

Window sills. Children do not climb on sills and do not lean out of windows to place the box. If the sill is high, the teacher makes the measurement and the children read the screen.

Cables. A USB cable stretched from the laptop to a distant surface is a trip hazard and spoils the measurement besides. WiFi solves both.

Books placed under edges are arranged so that nothing can slide off and fall on feet.

Adaptations for pupils with special educational needs

Colour only, no numbers. This is the main adaptation and it is not a concession: the child plays and ranks the surfaces by watching *the colour of the circle* and how quickly it turns green, never reading roll and pitch. The objective — that surfaces differ and that an instrument can say so — is met in full with no signed numbers at all.

The challenge, with support. For children with tremor or difficult fine control, the challenge is done with the box **already set down** on the desk, and their task is to *not touch the table* for ten seconds and to announce when the circle turns green. It is a real task and it can be won.

Three real roles for children with motor difficulties. The roles rotate at every surface:

- **the hands keeper** — checks before each attempt that nobody is touching the table and that the cable is slack, and **has the right to void an attempt**; this is the most valuable role in the group, because they decide which attempt is fair;
- **the screen reader** — says the two numbers, or just the colour;
- **the timekeeper** — counts the ten seconds aloud.

Frustration. A resetting timer can be hard to bear. **Announce the rule in advance** and say openly that most adults need several attempts — which is true. A child who knows failure is expected takes it entirely differently.

If things do not go to plan

Nobody can do the challenge. Check, in order: is the box on a surface? is the surface level — that is, does it read under 1° when nobody touches it? is the cable slack? **If the desk itself is tilted by four degrees the challenge is impossible**, however well a child holds it. Put a folded piece of paper under one leg, or move to the floor.

Everybody succeeds first time. The surface is very good and the group is calm — move to the eyes-closed extension, which is far harder and far more interesting.

The numbers never come near zero. If they stay stuck at a large value wherever you put the box, check that there is nothing under it and that the firmware is current — a box reading -1 g flat needs reflashing.

You have one box and thirty children. Screen projected, the challenge in turn, and **the roles from the adaptations box for everybody else** — the hands keeper, the screen reader and the timekeeper change at every attempt, so nobody is waiting idle.

The box will not connect. Do not stop the lesson. The tour of surfaces can be done with **a spirit level**, if you have one, and the discussion about the tilted sill and about pilots in cloud is unchanged. Measure again another day, after checking the cable, the port selected in the app and, if you work over WiFi, the network.

7. Further resources

Stable, well-established starting points. Search within each site for the topic named — these are base addresses, not deep links, so they do not rot.

- **NASA** — <https://www.nasa.gov> — search their pupil material on flight and cockpit instruments; suitable for the activity's final step.
- **Institute of Physics** — <https://www.iop.org> — search their primary material on forces and balance, written for teachers who are not specialists.
- **STEM Learning** — <https://www.stem.org.uk> — search for flight and balance for ages 7–11; many lesson plans this measurement slots straight into.
- **Royal Aeronautical Society** — <https://www.aerosociety.com> — material on the principles of flight, if the activity provokes questions about aircraft.
- **PhET Interactive Simulations** — <https://phet.colorado.edu> — search for the forces and balance simulations.
- **Skybrary** — <https://skybrary.aero> — for the teacher, on spatial disorientation in flight; it explains why the instrument beats the sensation.

8. Assessment

The exit ticket, identical to the one on the pupil sheet: *what did you change between attempts?* Collected in the last two minutes, it says more about the class’s understanding than any mark — and, in this activity, more than the number of attempts.

For a more formal judgement, use the **common three-axis rubric** — *measurement practice*, *data interpretation* and *scientific reasoning*. At Junior level the three axes look like this:

- **Measurement practice** — the child sets the box on the surface, places it with the same edge towards them every time, takes their hands off the table and leaves the cable slack.
- **Data interpretation** — the child says which surface is more level and by how much, and compares their ranking with the prediction.
- **Scientific reasoning** — the child describes an intervention — what they changed between attempts — and explains why it worked.

The rubric itself, with descriptors for all three levels, is defined once in the *curriculum plan* and is not repeated in the guides.

9. Pupil sheet

The sheet the children fill in is a separate document: `docs/en/student_sheets/flying_straight.pdf`. It has two pages — the prediction, the ranking of surfaces and their table on the first; the challenge, what I changed between attempts, and the conclusion on the second — so it is printed double-sided, one sheet per child.

Three things to know before printing it. **The sheet asks for the ranking *before* measuring**, as a prediction, and leaves room beside it for the real ranking — the collision between them is the lesson, and it has to be visible on the same page. **The challenge table asks for “what I changed”, not just the number of attempts**: without that column the game stays a game. **The number column has a “green / not” column beside it**, precisely for the child who does not work with signed numbers: they circle the second and leave the first empty. The tables are not repeated here precisely so that the two documents cannot drift apart: there is one printable version, and it is the sheet.